Java Practice Questions

1. Assignment operators
2. Leap year (with at least 3 conditions)
3. Bitwise operators
4. Odd or even
5. Logical operators
6. Check whether a number is divisible by both 5 and 11
7. Check whether a character is alphabet, digit, or special character
8. Check whether a number is multiple of 3 or 7
9. Check whether a number is palindrome
10. Check whether a number is Armstrong
11. Check whether a number is divisible by 3 without using %
12. Find maximum of two numbers without using if
13. Count even and odd numbers in an array
14. Count positive, negative, and zero elements
15. Print array elements in reverse order
16. Find the index of a given element
17. Left triangle pattern
18. Inverted left triangle pattern
19. Pyramid pattern
20. Diamond pattern
21. Sum of array elements in a 2D array
22. Getting input from the user in a 2D array
23. Strong number
24. Armstrong number
25. GCD of two numbers
26. LCM of two numbers
27. Butterfly pattern
28. Pascal pattern
29. Find the second largest element in an array
30. Remove duplicate elements from an array
31. Check whether a matrix is a square matrix
32. Check whether a matrix is an identity matrix
33. Print X pattern in a matrix
34. Print hollow square matrix

35. Palindrome using string (with equals() function)
36. Palindrome using string (without equals() function)
37. Count vowels and consonants using string
38. Count digits, alphabets, and special characters using string
39. Count number of words in a string
40. Remove spaces from a string

Java OOP Questions

Class & Objects

41. Create a Student class with id, name, and age. Create an object and display the details.
42. Create an Employee class with id, name, and salary. Display employee details.
43. Create a Rectangle class and calculate its area using an object.
44. Create a BankAccount class with accountNumber, holderName, and balance.
45. Create a Car class with brand, model, and price. Create three objects and display them.
46. Create a Student class with marks in three subjects and calculate total and average.
47. Create a Product class and find the most expensive product among three objects.
48. Create a Circle class and calculate area and circumference.
49. Create a Book class with title, author, and price. Create multiple objects.

Inheritance

50. Create an Animal class and inherit Dog from it.
51. Create a Person class and inherit Student from it.
52. Create a Vehicle class and inherit Car from it.
53. Create an Employee class and inherit Manager from it.
54. Create Animal → Dog → Puppy using multilevel inheritance.
55. Create Person → Employee → Manager using multilevel inheritance.
56. Create Animal → Dog, Cat using hierarchical inheritance.
57. Create Employee → Developer, Tester using hierarchical inheritance.
58. Create Vehicle → Car, Bike, Bus using hierarchical inheritance.
59. Create a BankAccount → SavingsAccount inheritance structure.

Polymorphism — Method Overloading

60. Create a Calculator class with overloaded add() methods.
61. Create a Calculator class with overloaded multiply() methods.
62. Create a Printer class with overloaded print() methods.
63. Create a Shape class with overloaded area() methods.
64. Create a Student class with overloaded display() methods.
65. Create a Payment class with overloaded pay() methods.

66. Create a Search class with overloaded search() methods.

Polymorphism — Method Overriding

67. Create an Animal class with sound() and override it in Dog.
68. Create an Animal class with sound() and override it in Cat.
69. Create a Vehicle class with start() and override it in Car.
70. Create an Employee class with calculateSalary() and override it in Manager.
71. Create a Shape class with area() and override it in Circle.
72. Create a Shape class with area() and override it in Rectangle.
73. Create a Payment class with makePayment() and override it in UPI and Card.
74. Create a BankAccount class with calculateInterest() and override it in SavingsAccount.